

Super-Gaussian: Multi-Contrast MRI Super-Resolution with Superpixel-Guided 2D Gaussian Tokenization

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Abstract—Reference-guided super-resolution (Ref-SR) methods can accelerate multi-contrast MRI (MC-MRI) acquisition, but their reliance on rigid patch-wise attention often fails to capture continuous anatomical structures, causing inaccurate reconstruction at ambiguous tissue boundaries. To address this, we propose Super-Gaussian (SG), a novel framework bridging discrete superpixel clustering with continuous Gaussian distributions. Specifically, SG introduces a “Gaussian Relaxation” mechanism, mapping the statistical attributes of deterministic superpixels to the parameters of probabilistic 2D Gaussian tokens. This mechanism enables each pixel to be influenced by overlapping Gaussians, effectively modeling smooth signal transitions and partial volume effects. We further incorporate a parallel attention scheme that synergizes fine-grained intra-superpixel refinement with global inter-Gaussian token interaction, allowing content-aware geometric updates of Gaussian tokens unlike fixed patches. Extensive experiments demonstrate that SG surpasses state-of-the-art methods in both reconstruction and downstream segmentation accuracy, delivering more clinically faithful MC-MRI results. Code is at <https://github.com/YC-Markus/SG>.

Index Terms—MRI super-resolution, multi-contrast, 2D Gaussian splatting, superpixel, image tokenization.

I. INTRODUCTION

MAGNETIC Resonance Imaging (MRI) is indispensable in clinical diagnosis due to its superior soft tissue contrast [1], [2]. Multi-contrast MRI (MC-MRI) further enhances diagnostic capabilities by providing complementary information across different imaging protocols. However, constrained by the fundamental physics of signal acquisition, obtaining certain modalities often necessitates significantly longer scanning duration. This limitation leads to reduced patient throughput, increased discomfort, and a heightened risk of motion artifacts. To address this dilemma, reference-guided super-resolution (Ref-SR) has emerged as a promising solution [2]–[10], which reconstructs high-resolution (HR) images for slow-acquisition modalities (e.g., T2w and FLAIR) by leveraging high-quality references from fast-acquisition contrasts (e.g., T1w).

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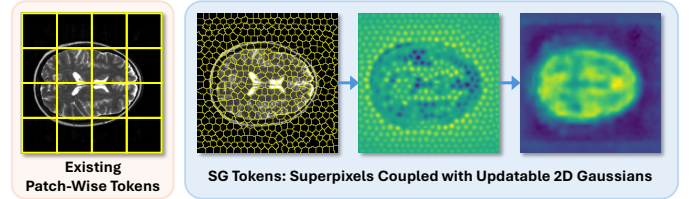


Fig. 1. A schematic comparison between the widely used patch-wise tokenization and the proposed SG tokenization. The former is extensively employed in super-resolution models like SwinIR [11], whereas the latter features continuous representation and content-aware capabilities.

In the realm of deep learning, patch-based transformers [11] are renowned for their long-range dependency modeling and have been adopted as backbones in the MC-MRI SR task [3], [5], [6], [8]. However, as illustrated in Fig. 1, these methods predominantly operate on fixed local neighborhoods like square patches, which rigidly partition the irregular tissue boundaries inherent in MR images. While subsequent token-based approaches in natural image SR [12], [13] attempt to cluster pixels into content-aware regions, they typically generate deterministic and fixed segmentation results. This discrete perspective conflicts with the continuous nature of biological tissues and the partial volume effects (PVE) common in MRI [14], introducing artificial discontinuities and failing to model the inherent signal ambiguity. Furthermore, the geometry of local regions is typically frozen after initialization. This lack of spatial flexibility often causes errors from the initial partitioning to propagate through the network.

More recently, 2D Gaussian splatting has been applied to various image tasks [15]–[18], demonstrating the superiority of continuous representations. Inspired by it, we propose Super-Gaussian (SG), a novel SR framework tailored to MC-MRI. As shown in Fig. 1, the core idea of SG lies in adopting discrete superpixels as a prior for initializing continuous Gaussian tokens, allowing the two representations to complement each other. Specifically, SG introduces a mechanism termed “Gaussian Relaxation”: by calculating the mean and covariance of

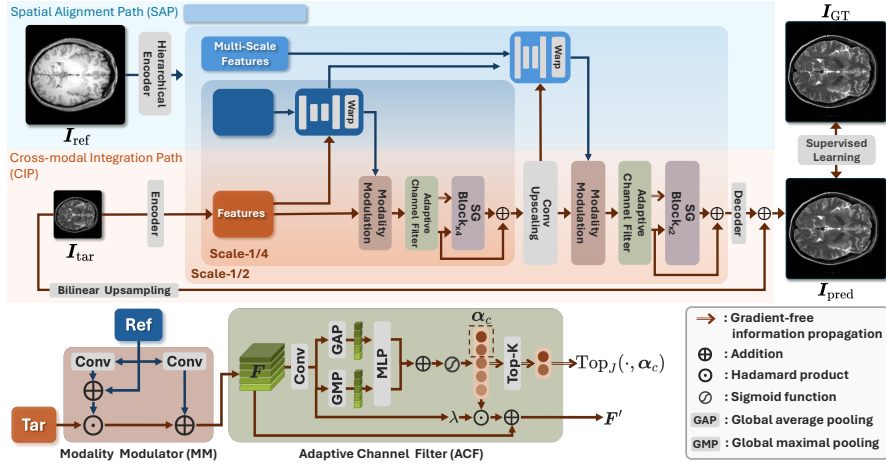


Fig. 2. An overview of the SG framework architecture. The detailed internal structure of SG Block is illustrated in Fig. 3.

each clustered superpixel, it transforms deterministic, fixed superpixels into probabilistic, geometry-adaptive 2D Gaussian tokens. This hybrid tokenization offers three fundamental advantages: 1) It allows each pixel to be influenced by multiple overlapping Gaussians, naturally modeling the ambiguous boundaries and signal mixing in MRI; 2) Unlike fixed patches, the feature vectors and geometric parameters of Gaussian tokens are dynamically updatable. This enables more accurate regional feature aggregation and spatial misalignment correction; 3) Previous work has noted the instability inherent in Gaussian parameter optimization [18]. Superpixels address it by constraining the Gaussian parameter space.

Building on this hybrid tokenization, SG incorporates a parallel attention scheme comprising Intra-Superpixel Attention (ISA) for fine-grained detail refinement, and Inter-Gaussian Token Interaction (IGTI) for global context modeling. Benefiting from the differentiable nature of 2D Gaussian splatting, the updated Gaussian tokens can propagate their feature vectors back to the pixel grid via a lightweight splatting operator. Furthermore, to accommodate the resolution gap and potential inter-modal misregistration commonly found in MC-MRI [5], [7], [10], we employ a hierarchical architecture for coarse-to-fine registration and fusion. Finally, to ensure computational efficiency, we implement custom GPU-parallelized kernels for superpixel clustering and 2D Gaussian splatting using Triton, effectively alleviating the possible performance bottlenecks.

Our main contributions are summarized as follows:

- We propose Super-Gaussian (SG), a MC-MRI SR framework that transcends rigid grid partitioning. Driven by the “Gaussian Relaxation” mechanism, it couples discrete superpixel clustering with continuous 2D Gaussian tokenization, modeling the smooth intensity transitions and fuzzy boundaries in MRI.
- We introduce a parallel attention scheme. It allows the geometric attributes of Gaussians and pixel-wise features to be dynamically updated, ensuring both global anatomical consistency and local detail fidelity.

- Extensive experiments on multiple datasets demonstrate that SG achieves state-of-the-art performance in both reconstruction and downstream tumor segmentation accuracy, producing clinically more faithful images.

II. METHOD

A. Overview of the Super-Gaussian Framework

The proposed SG framework is designed to reconstruct an HR target modality I_{GT} from an LR input I_{tar} guided by an HR reference I_{ref} . To address the significant resolution gaps and potential inter-modal misalignment common in MC-MRI, we employ a coarse-to-fine architecture spanning L scales. As illustrated in Fig. 2, SG framework comprises two coupled pathways: hierarchical Spatial Alignment Path (SAP) and Cross-modal Integration Path (CIP). Initially, we employ CNN-based extractors to obtain shallow features for both modalities, yielding single-scale features F_{tar}^0 for the target and multi-scale features $\{F_{ref}^l\}_{l=0}^L$ for the reference.

1) *Spatial Alignment Path*: To eliminate the potential impact of spatial misalignment between different modalities on subsequent fusion, we designed the SAP to pre-register the reference modality toward target modality at each resolution level. Specifically, at scale l , a lightweight registration network $\mathcal{R}^l$ estimates a residual displacement field based on the features of two modalities and the coarse-scale field U^{l-1} (for $l > 0$) from the preceding stage:

$$U^l = \mathcal{B}(U^{l-1}) + \mathcal{R}^l([F_{tar}^l, F_{ref}^l, \mathcal{B}(U^{l-1})]) \quad (1)$$

where $\mathcal{B}(\cdot)$ indicates bilinear interpolation. This field is employed to warp the reference features: $F_{ref}^{l'} = \text{Warp}(F_{ref}^l, U^l)$.

2) *Cross-Modal Integration Path*: Most existing approaches pre-upsample the LR input I_{tar} to the resolution of the reference I_{ref} before modality fusion. However, this strategy often introduces interpolation artifacts and imposes an unnecessary computational burden. In contrast, CIP reconstructs the target image from its original resolution.

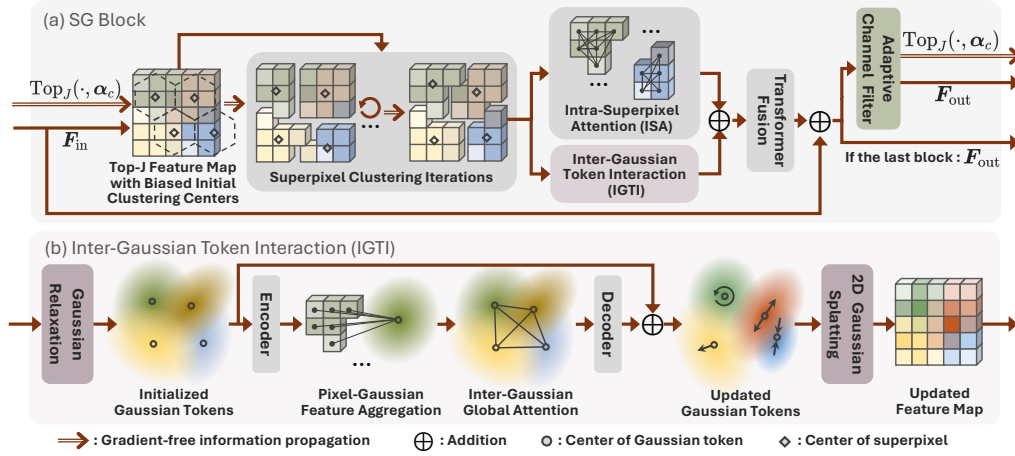


Fig. 3. A schematic diagram of proposed SG Block, detailing the regional tokenization and feature updating process facilitated by superpixels and 2D Gaussians. The update of a Gaussian primitive manifests as positional, geometric displacement, and the modification of its feature vector.

At each scale, to ensure the target modality retains fidelity while benefiting from reference guidance, a Modality Modulator (MM) is introduced. Unlike direct concatenation or addition, MM predicts the affine transformation parameters (scale γ and shift β) from the reference features $\mathbf{F}_{\text{ref}}^{l'}$ to modulate the target features $\mathbf{F}_{\text{tar}}^l$:

$$\mathbf{F}_{\text{in}}^l = \left(1 + \gamma(\mathbf{F}_{\text{ref}}^{l'})\right) \odot \mathbf{F}_{\text{tar}}^l + \beta(\mathbf{F}_{\text{ref}}^{l'}) \quad (2)$$

Following modulation, the features pass through an Adaptive Channel Filter (ACF) and a cascaded sequence of Super-Gaussian Blocks (detailed in section II-B). Finally, the fused features are upscaled or decoded to generate the residual image, optimized end-to-end with a standard $\mathcal{L}_1$ loss between $\mathbf{I}_{\text{pred}}$ and $\mathbf{I}_{\text{GT}}$: $\mathcal{L} = |\mathbf{I}_{\text{pred}} - \mathbf{I}_{\text{GT}}|$.

B. Hybrid Discrete-Continuous Tokenization

Discrete tokenization imposes rigid boundaries on soft tissues, while continuous tokens without prior constraint often suffer from optimization instability [18]. To bridge this gap, we propose a modified superpixel clustering method tailored to Gaussian initialization, and a ‘‘Gaussian Relaxation’’ mechanism to realize stable hybrid tokenization.

1) *Modified Superpixel Clustering*: Distinct from conventional superpixels, the clustering in SG operates within feature spaces. Normally, this introduces the ‘‘curse of dimensionality’’, where distance metrics become unreliable in high-dimensional spaces. To mitigate this, we propose an Adaptive Channel Filter (ACF). As shown in Fig. 2, ACF computes channel attention weights α_c via global average and max pooling [19], and then selects the Top- J most discriminative channels (usually set to 8) as candidates for distance calculation. Crucially, these weights are subsequently reinjected into main flow as a residual modulation, obviating the need for additional optimization objectives.

Furthermore, to better approximate the elliptical geometry inherent to Gaussian distributions, we initialize cluster centers on a hexagonal grid rather than a rectangular one, as shown

in Fig. 3(a). Finally, we leverage SLICO, a parameter-free variant of SLIC [20] implemented with custom Triton kernels, to achieve efficient and texture-aware clustering.

2) *Gaussian Relaxation Mechanism*: A recent study demonstrates that directly optimizing 2D Gaussian parameters from random initialization is ill-posed [18]. Fortunately, superpixels naturally contain the statistical priors needed to shrink the parameter space. Specifically, for each superpixel cluster $\mathcal{S}_k$ among K clusters, it comprises n_k pixels $\{\mathbf{p}_i\}_{i=1}^{n_k}$ with features $\{\mathbf{f}_i\}_{i=1}^{n_k}$. The Gaussian Relaxation transforms it into a probabilistic Gaussian token $\mathcal{G}_k(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k, \mathbf{v}_k)$ by explicitly mapping pixels’ statistical attributes as follows:

$$\begin{aligned} \boldsymbol{\mu}_k &= \frac{1}{n_k} \sum_{i=1}^{n_k} \mathbf{p}_i, & \mathbf{v}_k &= \frac{1}{n_k} \sum_{i=1}^{n_k} \mathbf{f}_i, \\ \boldsymbol{\Sigma}_k &= \frac{1}{n_k} \sum_{i=1}^{n_k} (\mathbf{p}_i - \boldsymbol{\mu}_k)(\mathbf{p}_i - \boldsymbol{\mu}_k)^\top, \end{aligned} \quad (3)$$

where $\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k, \mathbf{v}_k$ indicate the spatial center, the covariance matrix, and the feature vector of a 2D Gaussian. This initialization provides a robust starting point, facilitating the subsequent network to refine Gaussian tokens’ geometry and content.

C. Super-Gaussian Block

The SG Block employs a parallel attention scheme to update the hybrid tokens: Intra-Superpixel Attention (ISA) refines local details within discrete clusters, while Inter-Gaussian Token Interaction (IGTI) captures global context dependencies.

1) *Intra-Superpixel Attention*: To capture fine-grained textures, we treat pixels within each superpixel as variable-length sequences. As illustrated in Fig. 3(a), to accommodate the irregular geometry of individual superpixels, we apply variable-length self-attention equipped with relative position encoding to refine the pixel-wise features $\{\mathbf{f}_i\}$.

$$\{\mathbf{f}_i'\}_{i=1}^{n_k} = \text{SelfAttn}(\{\mathbf{f}_i + \text{PE}(\mathbf{p}_i - \boldsymbol{\mu}_k)\}_{i=1}^{n_k}) \quad (4)$$

where PE encodes the spatial offset from the cluster center μ_k . The updated features are scattered back to their original locations to form an updated feature map $\mathbf{F}_{\text{intra}}$.

2) *Inter-Gaussian Token Interaction*: Complementary to ISA, IGTI is designed for global information exchanges and updates. As illustrated in Fig. 3(b), the heterogeneous geometric parameters (μ_k, Σ_k) and feature v_k of each Gaussian are first projected into a homogeneous latent vector z_k via MLPs:

$$z_k = \mathcal{M}_{\mu}^{\text{enc}}(\mu_k) + \mathcal{M}_{\Sigma}^{\text{enc}}(\Sigma_k^{\text{para}}) + v_k \quad (5)$$

To enhance the regional representation capacity, we then perform pixel-to-Gaussian cross-attention, where the Gaussian latent vector z_k serves as the query, and the constituent pixel features $\{f_i\}$ serve as keys and values. Notably, we encode the Mahalanobis distance $D_M(\mathcal{G}_k, \mathbf{p}_i)$ into the keys to make the aggregation geometry-aware:

$$z'_k = z_k + \text{CrossAttn}(\mathbf{Q}_k, \{\mathbf{K}_i\}_{i=1}^{n_k}, \{\mathbf{V}_i\}_{i=1}^{n_k}) \quad (6)$$

After that, the aggregated Gaussian tokens undergo global self-attention to efficiently exchange long-range information:

$$\{z'_k\} = \text{SelfAttn}(\{z'_k\}_{k=1}^K) \quad (7)$$

The refined latent vectors are then decoded to predict residual updates for Gaussian parameters, forming the updated Gaussian tokens $\mathcal{G}_k(\mu''_k, \Sigma''_k, v''_k)$. Finally, $\{\mathcal{G}_k\}$ are rendered back to the pixel space as an updated feature map $\mathbf{F}_{\text{inter}}$. The feature value of pixel $\mathbf{p}$ is a weighted sum of overlapping Gaussians based on their probability density:

$$\mathbf{F}_{\text{inter}}(\mathbf{p}) = \sum_{k=1}^N v''_k \cdot \exp\left(-\frac{1}{2}D_M^2(\mathcal{G}_k, \mathbf{p})\right) \quad (8)$$

For computational efficiency, contributions from Gaussian tokens are truncated where the Mahalanobis distance exceeds a threshold of 3. We implement the rasterizer with custom Triton kernels, realizing batch-parallelized 2D Gaussian splatting.

3) *Auxiliary Components*: The final stage of the SG Block integrates the local and global features. The outputs from the parallel paths are summed as $\mathbf{F}_{\text{fused}} = \mathbf{F}_{\text{intra}} + \mathbf{F}_{\text{inter}}$. This fused feature is further processed by a transformer fusion module to enhance consistency. Then, we apply the channel attention weights α_c mentioned in section II-B1 as a residual modulation to produce the SG Block output $\mathbf{F}_{\text{out}}$.

III. EXPERIMENTS

A. Datasets

We evaluated our method on two public MC-MRI datasets:

1) *IXI Dataset*:¹We utilized 576 raw scans, using 256×256 T1w images to guide 64×64 T2w reconstruction. Slices with top 85% foreground ratios were selected, yielding approximately 51k / 6.3k / 6.3k slices for training / validation / testing.

2) *BraTS2021 Dataset* [21]: We utilized pre-registered scans with 240×240 T1w guiding 60×60 FLAIR reconstruction. Following the same foreground criterion, we obtained 133k / 16.6k / 16.6k splits for training / validation / testing.

¹<https://brain-development.org/ixi-dataset/>

TABLE I
QUANTITATIVE EVALUATION AND COMPUTATIONAL COST RESULTS ON IXI DATASET

Methods	Venues	IXI T1-T2			Computational Cost		
		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS ($\times 1e-2$) $\downarrow$	Params	FLOPs $\downarrow$	Infer Time (/slice) $\downarrow$
Bicubic	N/A	21.77	0.6937	24.46	N/A	N/A	N/A
MINet [4]	MICCAI'21	30.99	0.9322	5.900	6.90M	865.3G	75.7ms
WavTrans [3]	MICCAI'22	30.32	0.9248	6.448	2.11M	166.2G	126ms
McMRSR [8]	CVPR'22	29.76	0.9192	7.426	2.01M	173.8G	89.6ms
DCAMSR [6]	MICCAI'23	31.15	0.9341	5.657	9.29M	217.0G	19.4ms
FASR-Net [10]	TMM'23	30.02	0.9232	7.126	89.4M	742.2G	134ms
ECFNet [7]	ICPR'24	30.50	0.9272	6.287	11.1M	393.6G	38.4ms
SANet [9]	TNNLS'24	30.73	0.9292	6.244	4.78M	590.7G	58.7ms
A2-CDic [5]	TMI'25	30.71	0.9294	6.098	2.24M	905.9G	55.8ms
SG (ours)		31.67	0.9403	5.054	8.37M	98.3G	35.5ms

B. Baselines and Evaluation Metrics

We benchmarked against representative MC-MRI specific methods: MINet [4], WavTrans [3], McMRSR [8], DCAMSR [6], FASR-Net [10], ECFNet [7], SANet [9], A2-CDic [5]. Reconstruction performance is measured by PSNR, SSIM, and LPIPS. To further evaluate the clinical utility of the reconstructed images, we introduced a downstream tumor segmentation task evaluated via the Dice Similarity Coefficient (DSC) and Relative Volume Difference (RVD). These metrics assess the overlap accuracy and volumetric deviation between the segmentation masks derived from the super-resolved images and the ground truth.

C. Implementation Details

1) *Data Pre-processing*: For all datasets, each volume was normalized to the $[0, 1]$ range after clipping voxel intensities to the 99.5th percentile. LR inputs were synthesized via k-space truncation with a scaling factor of $4\times$. During training, data augmentation techniques were applied for all methods, including random 90-degree rotations and flips.

2) *Model Configuration*: To strike a balance between performance and computational cost, we utilized a two-level resolution hierarchy ($L = 2$). The first stage, operating at $1/4$ resolution, comprises 4 SG Blocks with a channel dimension of 128. The second stage at $1/2$ resolution consists of 2 SG Blocks with 96 channels. Derived by minimizing the theoretical computational complexity, the token counts K for SG Blocks in these stages were set to 256 and 512, respectively. All models were trained for 650,000 steps with a batch size of 4 on NVIDIA RTX A6000 GPU using PyTorch.

D. Comparison Results with State-of-the-art

1) *Reconstruction Performance & Computational Cost*: We compared the proposed SG with SOTA methods on both the IXI and BraTS2021 datasets. As presented in Table I, SG outperforms all competing methods, achieving a PSNR gain ranging from 0.5 dB to 1.9 dB, along with a reduction of at least 10% in LPIPS perceptual loss. This is qualitatively corroborated by the visual comparisons in Fig. 4(a). As evidenced by the lowest residual intensities relative to the ground truth, SG recovers the most accurate tissue boundaries between gray matter and white matter. Similar trends are

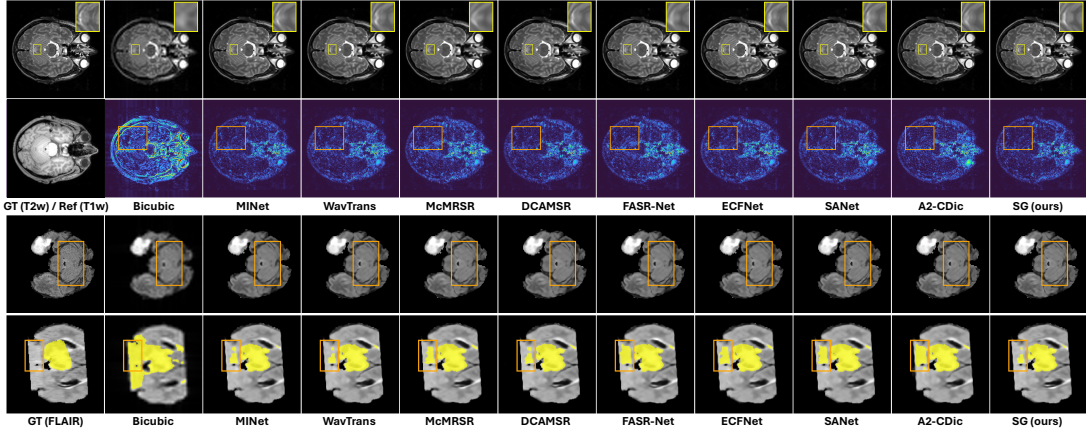


Fig. 4. Representative reconstruction results from (a) IXI dataset and (b) BraTS2021 dataset. The second row illustrates the reference T1w image, and the differences between reconstruction results and the ground truth. The fourth row illustrates the whole tumor segmentation masks predicted by pre-trained segmentation model using the reconstructed images from different MC-MRI SR methods.

TABLE II
QUANTITATIVE RECONSTRUCTION AND DOWNSTREAM SEGMENTATION
RESULTS ON BRA-TS2021 DATASET

Methods	BraTS2021 T1-FLAIR			Segmentation Results	
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS ($\times 1e-2$) $\downarrow$	DSC $\uparrow$	RVD $\downarrow$
Bicubic	27.75	0.7981	18.86	0.8679	24.4%
MINet	38.44	0.9683	2.888	0.9622	6.95%
WavTrans	38.08	0.9667	3.180	0.9612	7.01%
McMRSR	38.31	0.9676	3.184	0.9585	8.15%
DCAMSR	38.43	0.9686	2.871	0.9626	6.86%
FASR-Net	37.60	0.9644	3.862	0.9556	9.08%
ECFNet	37.96	0.9663	3.182	0.9597	7.64%
SANet	38.40	0.9678	3.029	0.9605	7.43%
A2-CDic	38.46	0.9682	3.104	0.9565	8.65%
SG (ours)	39.06	0.9712	2.658	0.9642	6.70%

observed on BraTS2021 dataset as shown in Table II. Besides, benefiting from efficient image tokenization, SG also exhibits the lowest FLOPs and the second-fastest inference speed, thereby facilitating its deployment in practical applications.

2) *Downstream Segmentation Performance*: To validate the practical feasibility of our method in clinical workflows, we conducted downstream segmentation experiments on BraTS dataset. We trained a 3D SegResNet [22] model on the original HR T1w and FLAIR modalities to segment the whole tumor. During testing, the FLAIR images generated by different SR methods were fed into the fixed segmentation network alongside T1w reference. As reported in Table II, SG achieves the highest scores in DSC and the lowest errors in RVD. Also in Fig. 4(b), the segmentation masks generated from SG-reconstructed images align most closely with the ground truth, preserving precise tumor boundaries. These results confirm that SG not only restores visual fidelity, but also preserves critical semantic information required for clinical analysis.

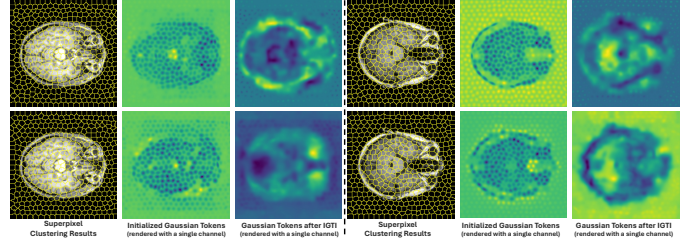


Fig. 5. Visualization of SG tokens for two samples in different SG Blocks. It illustrates Gaussian tokens' capability to globally aggregate content-aware information, such as the skull, eyeballs, and specific white matter regions.

TABLE III
ABLATION STUDY RESULTS FOR SG FRAMEWORK ON IXI DATASET

Submodule Ablation	IXI T1-T2			Computational Cost		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Params	FLOPs $\downarrow$	Infer Time (/slice) $\downarrow$
w/o Spatial Alignment	31.55	0.9392	5.180	7.30M	87.4G	34.3ms
w/o Hierarchical Design	31.17	0.9352	5.659	7.01M	67.8G	13.9ms
w/o Superpixel Clustering	31.56	0.9390	5.281	8.37M	98.3G	27.3ms
w/o ACF	31.59	0.9394	5.168	8.23M	96.7G	33.7ms
w/o ISA	31.55	0.9390	5.192	7.69M	89.5G	30.4ms
w/o IGTI	31.51	0.9386	5.220	6.24M	93.6G	26.9ms
Complete Version	31.67	0.9403	5.054	8.37M	98.3G	35.5ms

E. Visualization of SG Tokenization Process

Fig. 5 illustrates the results of SG tokenization in different SG Blocks. It is evident that superpixels provide effective initialization priors for Gaussian tokenization. Following the IGTI module, the Gaussian tokens exhibit a distinct attentional preference for specific anatomical regions, demonstrating their capability for content-aware global information aggregation.

F. Ablation Studies

To investigate the contribution of individual components in SG framework, we conducted comprehensive ablation studies. As summarized in Table III, we evaluated the model's behavior

by isolating specific modules, measuring both reconstruction quality and computational cost.

1) *Impact of Hierarchical and Alignment Design:* First, we examined the macro-architectural designs. Removing the Spatial Alignment Path (SAP) resulted in a performance drop of 0.12 dB, confirming that explicit feature-level registration is crucial for correcting spatial discrepancies in MC-MRI. Furthermore, excluding the fine-grained processing stage (w/o Scale-1/2 Stage) led to a significant degradation in detail recovery, validating the necessity of coarse-to-fine processing.

2) *Impact of Superpixel-based Tokenization:* To validate the effectiveness of our hybrid tokenization, we replaced the superpixel clustering with standard rectangular patches. This modification caused a notable decline in PSNR (0.11 dB), supporting our hypothesis that rigid grid partitioning conflicts with continuous tissue structures. Additionally, removing the Adaptive Channel Filter (ACF) also negatively impacted performance, demonstrating its importance in mitigating the “curse of dimensionality” during feature-space clustering.

3) *Effectiveness and Efficiency of Parallel Attention:* Finally, we analyzed the specific contributions of the parallel attention mechanisms within the SG Block. We observe that both ISA and IGTI contribute significantly to performance gains (improving 0.12 dB and 0.16 dB, respectively) while incurring minor computational cost. This confirms that our parallel attention scheme efficiently balances fine-grained local refinement with global context modeling without imposing a heavy computational burden.

IV. CONCLUSION

In this paper, we proposed Super-Gaussian (SG), a novel framework for MC-MRI SR that overcomes the limitations of rigid grid partitioning. By bridging discrete superpixel clustering with continuous 2D Gaussian tokenization via “Gaussian Relaxation” mechanism, SG effectively models the continuous anatomical structures inherent in MRI. We further introduced a parallel attention scheme to dynamically refine both local details and global content-aware dependencies. Extensive experiments on the IXI and BraTS2021 datasets demonstrate that SG achieves state-of-the-art performance in both reconstruction quality and downstream tumor segmentation accuracy.

REFERENCES

- [1] Ivo G Schoots, Hashim U Ahmed, Peter Albers, Patrick Asbach, Roderick CN van den Bergh, Rebecka A Godtman, Pim J van Leeuwen, Tobias Nordström, Shonit Punwani, Jonas Wallström, et al., “Magnetic resonance imaging-based biopsy strategies in prostate cancer screening: A systematic review,” *European Urology*, 2025.
- [2] Hanguang Xiao, Zhiying Yang, Tianqi Liu, Shihong Liu, Xiaoxuan Huang, and Jiahui Dai, “Deep learning for medical imaging super-resolution: A comprehensive review,” *Neurocomputing*, p. 129667, 2025.
- [3] Guangyuan Li, Jun Lyu, Chengyan Wang, Qi Dou, and Jing Qin, “Wav-trans: Synergizing wavelet and cross-attention transformer for multi-contrast mri super-resolution,” in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2022, pp. 463–473.
- [4] Chun-Mei Feng, Huazhu Fu, Shuhao Yuan, and Yong Xu, “Multi-contrast mri super-resolution via a multi-stage integration network,” in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2021, pp. 140–149.
- [5] Pengcheng Lei, Miaomiao Zhang, Faming Fang, and Guixu Zhang, “Robust deep convolutional dictionary model with alignment assistance for multi-contrast mri super-resolution,” *IEEE Transactions on Medical Imaging*, 2025.
- [6] Shoujin Huang, Jingyu Li, Lifeng Mei, Tan Zhang, Ziran Chen, Yu Dong, Linzheng Dong, Shaojun Liu, and Mengye Lyu, “Accurate multi-contrast mri super-resolution via a dual cross-attention transformer network,” in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2023, pp. 313–322.
- [7] Zhiyuan Yang, Bo Zhang, Zhiqiang Zeng, and Si Yong Yeo, “Edge-guided and cross-scale feature fusion network for efficient multi-contrast mri super-resolution,” in *International Conference on Pattern Recognition*. Springer, 2024, pp. 208–218.
- [8] Guangyuan Li, Jun Lv, Yapeng Tian, Qi Dou, Chengyan Wang, Chenliang Xu, and Jing Qin, “Transformer-empowered multi-scale contextual matching and aggregation for multi-contrast mri super-resolution,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 20636–20645.
- [9] Chun-Mei Feng, Yunlu Yan, Kai Yu, Yong Xu, Huazhu Fu, Jian Yang, and Ling Shao, “Exploring separable attention for multi-contrast mr image super-resolution,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 9, pp. 12251–12262, 2024.
- [10] Yiming Liu, Mengxi Zhang, Bo Jiang, Bo Hou, Dan Liu, Jie Chen, and Heqing Lian, “Flexible alignment super-resolution network for multi-contrast magnetic resonance imaging,” *IEEE Transactions on Multimedia*, vol. 26, pp. 5159–5169, 2023.
- [11] Jingyun Liang, Jie Zhang Cao, Guolei Sun, Kai Zhang, Luc Van Gool, and Radu Timofte, “Swinir: Image restoration using swin transformer,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 1833–1844.
- [12] Aiping Zhang, Wenqi Ren, Yi Liu, and Xiaochun Cao, “Lightweight image super-resolution with superpixel token interaction,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 12728–12737.
- [13] Xin Liu, Jie Liu, Jie Tang, and Gangshan Wu, “Catanet: Efficient content-aware token aggregation for lightweight image super-resolution,” in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 17902–17912.
- [14] Yang Yang, Jiaxin Zhuang, Guoying Sun, Ruixuan Wang, and Jingyong Su, “Boundary-guided contrastive learning for semi-supervised medical image segmentation,” *IEEE Transactions on Medical Imaging*, 2025.
- [15] Zhengqiang Zhang, Rongyuan Wu, Lingchen Sun, and Lei Zhang, “Gpstocken: Gaussian parameterized spatially-adaptive tokenization for image representation and generation,” *arXiv preprint arXiv:2509.01109*, 2025.
- [16] Jiajun Dong, Chengkun Wang, Wenzhao Zheng, Lei Chen, Jiwen Lu, and Yansong Tang, “Gaussiantoken: An effective image tokenizer with 2d gaussian splatting,” *arXiv preprint arXiv:2501.15619*, 2025.
- [17] Yunxiang Zhang, Bingxuan Li, Alexandr Kuznetsov, Akshay Jindal, Stavros Diolatzis, Kenneth Chen, Anton Sochenov, Anton Kaplanyan, and Qi Sun, “Image-gs: Content-adaptive image representation via 2d gaussians,” in *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers*, 2025, pp. 1–11.
- [18] Long Peng, Anran Wu, Wenbo Li, Peizhe Xia, Xueyuan Dai, Xinjie Zhang, Xin Di, Haoze Sun, Renjing Pei, Yang Wang, et al., “Pixel to gaussian: Ultra-fast continuous super-resolution with 2d gaussian modeling,” *arXiv preprint arXiv:2503.06617*, 2025.
- [19] Sanghyun Woo, Jongchan Park, Joon-Young Lee, and In So Kweon, “Cbam: Convolutional block attention module,” in *Proceedings of the European conference on computer vision (ECCV)*, 2018, pp. 3–19.
- [20] Radhakrishna Achanta, Appu Shaji, Kevin Smith, Aurelien Lucchi, Pascal Fua, and Sabine Süsstrunk, “Slic superpixels compared to state-of-the-art superpixel methods,” *IEEE transactions on pattern analysis and machine intelligence*, vol. 34, no. 11, pp. 2274–2282, 2012.
- [21] Bjoern H Menze, Andras Jakab, Stefan Bauer, Jayashree Kalpathy-Cramer, Keyvan Farahani, Justin Kirby, Yuliya Burren, Nicole Porz, Johannes Slotboom, Roland Wiest, et al., “The multimodal brain tumor image segmentation benchmark (brats),” *IEEE transactions on medical imaging*, vol. 34, no. 10, pp. 1993–2024, 2014.
- [22] Andriy Myronenko, “3d mri brain tumor segmentation using autoencoder regularization,” in *International MICCAI brainlesion workshop*. Springer, 2018, pp. 311–320.